

Course: BTech

Semester: 6

Prerequisite: Parallel Computing, Computer Architecture | 303105210 - Computer Organization and Microprocessor

Rationale : HPC is more than just for achieving high performance - it is a compelling vision for how computation can seamlessly scale from a single processor to virtually limitless computing power. The market demands general-purpose processors that deliver high single threaded performance as well as multi-core throughput for a wide variety of workloads on client, server, and high- performance computing (HPC) systems.

Teaching and Examination Scheme										
Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	0	3	20	20	-	60	-	100

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content W - Weightage (%), T - Teaching hours

Sr.	Topics	W	T
1	Introduction to Parallel Computing Introduction to Parallel Computing: Motivating Parallelism, Scope of Parallel Computing, Organization and Contents of the Text, Parallel Programming Platforms: Implicit Parallelism: Trends in Microprocessor & Architectures, Limitations of Memory System Performance, Dichotomy of Parallel Computing Platforms, Physical Organization of Parallel Platforms, Communication Costs in Parallel Machines Levels of parallelism (instruction, transaction, task, thread, memory, function) Models (SIMD, MIMD, SIMT, SPMD, Dataow Models, Demand-driven Computation) Architectures: N-wide superscalar architectures, multi-core, multi-threaded.	20	10
2	Parallel Algorithm Design Principles of Parallel Algorithm Design: Preliminaries, Decomposition Techniques, Characteristics of Tasks and Interactions, Mapping Techniques for Load Balancing, Methods for Containing Interaction Overheads, Parallel Algorithm Models.	15	8
3	Programming Using the Message-Passing Paradigm Principles of Message- Passing Programming, The Building Blocks: Send and Receive Operations, MPI: the Message Passing Interface, Topology and Embedding, Overlapping Communication with Computation, Collective communication, and Computation Operations	15	8
4	Synchronization Scheduling, Job Allocation, Job Partitioning, Dependency Analysis Mapping Parallel Algorithms onto Parallel Architectures, Thread Basics, The POSIX Thread API, Thread Basics: Creation and Termination, Synchronization Primitives in Pthreads, Controlling Thread and Synchronization Attributes, Thread Cancellation, Composite Synchronization Constructs, Tips for Designing Asynchronous Programs, OpenMP: a Standard for Directive Based Parallel Programming	20	9
5	Parallel Programming Using GPGPU An Overview of GPGPU, DGX architecture, An Overview of GPGPU Programming, An Overview of GPGPU Memory Hierarchy Features, CUDA Programming	15	5



6	Performance measures Speedup, efficiency, and scalability. Abstract performance metrics (work, critical paths), Amdahl's Law, abstract vs. Real performance (granularity, scalability)	15	5
Total		100	45

Reference Books

1.	Introduction to Parallel Computing By Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar Pearson Publication (TextBook)
2.	Advanced Computer Architecture: Parallelism, Scalability, Programmability By Kai Hwang, Naresh Jotwani, McGraw Hill, Second Edition, 2010
3.	CUDA by Example An Introduction to General Purpose GPU Programming By Edward Kandrot and Jason Sanders, Addison- Wesley Professional, 2010

Course Outcome

1. To understand the basic parallel programming concepts & GPU.
2. To comprehend the various High Performance Computing Paradigms.
3. To design parallel algorithms.
4. To demonstrate the Message passing and synchronization concepts.
5. To Perform basic CUDA Programming